

Modulation of angiogenesis by dietary phytoconstituents in the prevention and intervention of breast cancer.

Breast cancer is the leading cause of cancer-related deaths for women in the United States and the rest of the world. About 8% of women develop breast cancer during the course of their lives. Dietary habits are closely associated with both the risk and progression of breast cancer. Dietary agents have accumulated increasing importance with regards to the prevention and treatment of breast cancer. One such manner by which these compounds can target breast cancer development and progression is through interference with the angiogenic pathways. Angiogenesis is an intricate process that involves the development of new capillaries from previously existing blood vessels. Disruption of this pathway, therefore, provides a novel and effective avenue for therapeutic intervention of breast cancer. Various phytochemicals found in the diet kill breast cancer cells in vitro and prevent as well as suppress breast cancer progression in various preclinical animal models. This review examines the value of dietary phytoconstituents in the prevention and treatment of breast cancer through modulation of the intricate and complex process of angiogenesis. In addition, the potential benefits, challenges, and future directions of research on anti-angiogenic dietary phytochemicals in the prevention and intervention of breast cancer are also addressed. Copyright © 2012 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.